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Matplotlib & Seaborn Cheat Sheet

Data visualisation, plot types, styling & customisation.

FIGURE & AXES SETUP

```
fig, ax = plt.subplots()
fig, ax = plt.subplots(2,
fig, ax = plt.subplots(fi
fig.suptitle('Main Title'
plt.tight_layout()
plt.savefig('fig.png',dpi
plt.show()
```

LINE & SCATTER PLOTS

```
ax.plot(x, y)
ax.plot(x, y, 'r--', lw=2
ax.scatter(x, y, c='blue'
ax.scatter(x, y, c=z, cma
ax.plot(x, y, marker='o',

ax.plot(x, y1, label='A')
```

BAR & HISTOGRAM

```
ax.bar(x, height)
ax.barh(y, width)
ax.bar(x, y, color='steel
ax.hist(data, bins=30)
ax.hist(data, bins=30, de
ax.hist2d(x, y, bins=20)
ax.bar(x, y, bottom=base)
```

ANNOTATIONS & LABELS

```
ax.set_title('Title')
ax.set_xlabel('X Label')
ax.set_ylabel('Y Label')
ax.legend(loc='upper righ
ax.annotate('text',xy=(x,
ax.axhline(y=0, color='k'
ax.axvline(x=0, color='r'
```

AXIS CUSTOMISATION

```
ax.set_xlim(0, 10)
ax.set_ylim(-1, 1)
ax.set_xticks([0,2,4,6,8]
ax.set_xticklabels(['a','
ax.tick_params(axis='x',r
ax.invert_yaxis()
ax.set_aspect('equal')
```

STYLES & COLOUR

```
plt.style.use('seaborn-v0
plt.style.use('ggplot')
plt.style.use('dark_backg
cmap = plt.cm.viridis
colors = plt.cm.tab10.col
ax.set_facecolor('#f5f5f5
fig.patch.set_facecolor('
```

SEABORN DISTRIBUTION

```
sns.histplot(df, x='col')
sns.kdeplot(df, x='col')
sns.boxplot(df, x='cat',
sns.violinplot(data=df, x
sns.stripplot(data=df, x=
sns.ecdfplot(data=df, x='
sns.rugplot(data=df, x='c
```

SEABORN CATEGORICAL

```
sns.barplot(df, x='cat',
sns.countplot(df, x='cat'
sns.pointplot(df, x='cat'
sns.catplot(df, x='x', y=
  col='facet', kind='box'
sns.swarmplot(data=df, x=
```

SEABORN RELATIONAL

```
sns.scatterplot(df, x='a'
sns.lineplot(df, x='a', y
sns.relplot(df, x='a', y=
  col='facet', kind='scat

sns.pairplot(df)
sns.pairplot(df, hue='tar
```

SEABORN HEATMAPS

```
sns.heatmap(df.corr())
sns.heatmap(df.corr(), an
sns.heatmap(data, cmap='c
sns.heatmap(data, fmt='.2
sns.clustermap(df.corr())

sns.heatmap(data, linewid
```

SUBPLOTS & LAYOUT

```
fig, axes = plt.subplots(
axes[0,0].plot(x,y)
gs = fig.add_gridspec(2,3
ax = fig.add_subplot(gs[0
plt.subplot2grid((3,3),(0
  colspan=2)
fig.subplots_adjust(hspac
```

SAVING & EXPORT

```
plt.savefig('out.png', dp
plt.savefig('out.pdf')
plt.savefig('out.svg')

plt.savefig('out.png',
  bbox_inches='tight',
  transparent=True)
```